3 Out-of-Distribution (OOD) Classifier

In this section, we introduce three important components of our method: entropy based margin-loss function (3.1), training ensemble of leave-out classifiers (3.2), and OOD detection scores (3.3).

Algorithm 1: Algorithm to train K Leave Out Classifiers

Input : Training Data X, Number of classes N, K Partitions, δ accuracy bound, Validation OOD Data X_{valOOD}
Output : K Leave Out Classifiers

```
for i ← 1 to K do
    X_ood ← X_i, X_in ← X − X_i;
    while Not Converged do
        ood_batch ← Sample OOD minibatch;
        in_batch ← Sample in-distribution minibatch;
        update the classifier F_i by minimizing loss (Equation 1) using SGD;
        save model with least OOD error on X_valOOD within δ accuracy of
          current best accuracy.;
    end
end
return {F_i};
```

3.1 Entropy based Margin-Loss

Given a labeled set $(x_i \in X_{in}, y_i \in Y_{in})$ of in-distribution (ID) samples and $(x_o \in X_{ood})$ of out-of-distribution (OOD) samples, we propose a novel loss term in addition to the standard cross-entropy loss on ID samples. This loss term seeks to maintain a margin of at least m between the average entropy of OOD and ID samples. Formally, a multi-layer neural network $F : x \rightarrow p$ which maps an input x to probability over classes and parametrized by W is learned by minimizing the margin-loss over the difference of average entropies over OOD samples and ID samples, and cross entropy loss on ID samples. The loss function is given by Equation 1,

$$\mathcal{L} = -\frac{1}{|X_{in}|} \sum_{x_i \in X_{in}} \log(F_{y_i}(x_i)) + \beta * \max\left(m + \frac{\sum_{x_i \in X_{in}} H(F(x_i))}{|X_{in}|} - \frac{\sum_{x_o \in X_{ood}} H(F(x_o))}{|X_{ood}|}, 0\right) \quad (1)$$

where $F_{y_i}(x_i)$ is the predicted probability of sample x_i whose ground truth class y_i, $H(\cdot)$ is the entropy over the softmax distribution, m is the margin and β is the weight on margin entropy loss.

The new loss term evaluates to its minimum value *zero* when the difference between the average entropy of OOD and ID samples is greater than the margin m. For ID samples, the entropy loss encourages the softmax probabilities of

non ground-truth classes to decrease and the cross-entropy loss encourages the softmax probability of ground-truth class to increase. For OOD samples, the entropy loss encourages the probabilities of all the classes to be equal. When the OOD entropy is higher than ID entropy by a margin m, the new loss term evaluates to *zero*. Our experiments suggest that maximizing OOD entropy leads to overfitting. Bounding the difference of average entropies of ID samples and OOD samples entropies with margin has helped in preventing overfitting, and thus is better for model generalization[11].

Algorithm 2: Algorithm for OOD Detection using K Leave Out Classifiers

Input : Test Image x_t, K leave-out Classifiers $F_i, i \in 1, ..., K$ and their temperature scaled versions $F_i(x_t; T)$, perturbation factor ϵ, number of classes N

Output : Class prediction C_t, OOD score O_t

$S_t \leftarrow \{0\}^N$, $O_t \leftarrow 0$;
for $i \leftarrow 1$ ***to*** K **do**
  $S_t \leftarrow S_t + F_i(x_t)$;
  $\hat{x}_t \leftarrow x_t - \epsilon * \text{sign}(\frac{\partial H(F_i(\hat{x}_t;T))}{\partial x_t})$;
  $O_t \leftarrow O_t + \max_N(F_i(\hat{x}_t; T)) - H(F_i(\hat{x}_t; T))$;
end
$C_t \leftarrow \text{argmax}(S_t)$;
return C_t, O_t;

3.2 Training Ensemble of leave-out classifiers

Given an in-distribution training data X which consists of N classes, the data is divided into K partitions $X_i, i \in \{1, ..., K\}$ such that the classes of each partition are mutually exclusive to all other partitions. A set of K classifiers are learned where classifier $F_i, i \in \{1, ..., K\}$ uses the partition X_i as OOD data X_{ood} and rest of the data $X - X_i$ as in-distribution data X_{in}. A particularly simple way of partitioning the classes is to divide them into partitions with equal number of classes. For example, dividing a dataset of $N = 100$ classes into $K = 5$ random and equal partitions gives us a partition with size of 20 classes. Each of the K classifiers would then use 20 classes for OOD and 80 classes as ID. Each classifier F_i is learned by minimizing the proposed margin entropy loss (eqn 1) using the assigned OOD and ID data. During the training, we also assume a small number of out-of-distribution images to be available as a validation dataset. At every epoch, we save the model with best OOD detection rate on this small OOD validation data and within a δ accuracy bound of the current best accuracy. The complete algorithm for training the leave-out classifiers in presented in Algorithm 1.

3.3 OOD Detection Score for Test Image

At the time of testing, an input image is forward propagated through all the K leave-out classifiers and the softmax vectors of all the networks are remapped to their original class indices. For the left-out classes, a score of zero is assigned. For classification of an input sample, first the softmax vectors of all the classifiers are averaged and the class with the highest averaged softmax score is considered as the prediction. For the OOD detection score, for each of the K classifiers, we first compute both the maximum value and negative entropy of the softmax vectors with temperature scaling. We then compute the average of all these values to obtain the OOD detection score.

An in-distribution sample with class labels y_i acts as an OOD for exactly one of the K classifiers. This is because the classes are divided into K mutually exclusive partitions and class y_i can be part of only one of these partitions. When an in-distribution sample (x_i, y_i) is forward propagated through these K classifiers, we would expect the negative entropy and maximum softmax score to be high for $K-1$ classifiers where it was sampled as in-distribution dataset. However, for an OOD sample x_o we expect the negative entropy and maximum softmax score to be relatively low for all the K classifiers. We thus expect a higher OOD detection score for ID samples than the OOD samples thus differentiating them.

Following the work of ODIN [13], we use both temperature scaling and input preprocessing while testing. In temperature scaling, the logits feeding into softmax layer are scaled by a constant factor T. It has been established that temperature scaling can calibrate the classification score and in the context of OOD detection [13], it pushes the softmax scores of in- and out-of-distribution samples further apart when compared to plain softmax. We modify input preprocessing by perturbing over entropy loss instead of cross-entropy loss used by ODIN [13]. Perturbing using the entropy loss decreases the entropy of the ID samples much more than the OOD samples. For an input test image x_t, after it is forward propagated through the neural network F_i, the gradient of the entropy loss with respect to x_t is computed and the input is perturbed with Equation 2. The OOD detection score is then calculated by the combination of maximum softmax value and entropy as described previously, both with temperature scaling.

$$\hat{x_t} = x_t - \epsilon * \text{sign}(\frac{\partial L(F_i(x_t; T))}{\partial x_t}) \tag{2}$$

The complete algorithm for OOD detection on a test image is presented in Algorithm 2.

4 Experimental Results

In this section, we describe our experimental results. The details such as in-distribution and OOD datasets, the neural network architectures and evaluation metrics are described in detail. The ablation studies on various hyper-parameters